

BMW Used Cars Data Analysis & Cleaning Report

DATA QUALITY ASSESSMENT & FEATURE ENGINEERING REPORT

Executive Summary

The primary objective of this project was to transform a raw dataset of **3,060** BMW used car listings into a high-integrity, analysis-ready format. This process involved rigorous data cleaning, outlier mitigation, and feature engineering to ensure that subsequent Exploratory Data Analysis (EDA), Key Performance Indicator (KPI) tracking, and Power BI dashboard integrations are based on accurate metrics. The final cleaned dataset consists of **3,051** high-quality records.

3,051

CLEANED RECORDS

\$27,799

AVERAGE PRICE

121,608

AVERAGE MILEAGE

3 Series

TOP MODEL

1. Data Quality Assessment & Cleaning

A. Text Standardization & Formatting:

- Whitespace Removal:** Stripped all leading and trailing whitespaces across `model`, `transmission`, `fuelType`, and `color` features.
- Case Standardization:** Applied Title Case across categorical columns to ensure entries like `"automatic"` and `"Automatic"` are merged.
- Preservation of Technical Nomenclature:** Special logic was implemented to prevent standard casing from distorting specific electric brand names (e.g., `i3` and `i8` models).

B. Negative Price Correction & Invalid Data Handling:

- Negative Prices:** Identified 3 records with invalid negative price values (`price = -500.0`) in listings `BMW100675`, `BMW101886`, and `BMW102874`. These values were converted to `NaN` for statistical imputation.
- Invalid MPG Values:** Converted 3 records with non-positive fuel efficiency ratings (`mpg <= 0`) into `NaN`, as zero or negative fuel consumption is physically impossible.

C. Outlier Filtering & Record Removal:

To prevent extreme data points from skewing results, **9 records** were removed based on strict business logic:

- Future Manufacturing Years:** Excluded 3 listings (`BMW100178`, `BMW100273`, `BMW101590`) reporting a year of `2027`.
- Extreme Mileage Outliers:** Excluded 3 entries (`BMW100562`, `BMW100643`, `BMW101537`) containing placeholder values of `999,999 miles`.
- Extreme Price Outliers:** Filtered 3 listings (`BMW100192`, `BMW100539`, `BMW101093`) with synthetic prices of `$350,000`.

D. Imputation of Missing Values:

- **Numeric Columns:** Columns including price (30), mileage (46), tax (121), mpg (151), and engineSize (61) were filled using the column **Median** to minimize the impact of skewed distributions.
- **Categorical Columns:** For the color column, 187 missing values were imputed using the statistical **Mode** (Red).

2. Feature Engineering & Added Columns

Four new features were engineered to provide granular insights into vehicle value and usage patterns:

Feature Name	Logic / Calculation	Business Value to Client
car_age	2024 - year	Translates the manufacture year into age in years, essential for calculating depreciation curves.
price_per_mile	price / mileage	Provides a unit-cost efficiency metric to identify high-value deals regardless of total price.
is_luxury_model	Boolean flag (M-Series, X5-X7, 7/8 Series)	Enables immediate market segmentation to compare premium performance assets against standard models.
mileage_per_year	mileage / car_age	Distinguishes between heavy commercial usage and light personal usage to evaluate risk factors.

3. Summary KPIs & Business Insights

- **Pricing Overview:** The average listing price is approximately **\$27,799**, with a median price of **\$26,775**, indicating a balanced market with slight right-skewness.
- **Usage Overview:** The average mileage across the fleet stands at **121,608 miles**.
- **Model Leadership:** The **3 Series** remains the most prevalent model with 531 listings. Performance-tier models, specifically the **M2**, command the highest average valuation at approximately **\$62,813**.
- **Year vs. Price:** A strong positive correlation of **+0.66**, confirming that newer models significantly drive higher prices.
- **Mileage vs. Price:** A strong inverse correlation of **-0.60**, highlighting the impact of wear and tear on resale value.
- **Tax and MPG:** These factors demonstrated negligible direct linear impact on vehicle pricing within this specific dataset.